Complete binding for the actual F_5 BK certificate

Result. BKFullBindingAudit.full_bk_certificate writes the outer algebraically closed field L, Field/Algebra/DecidableEq assumptions and complete actual-object conclusion explicitly. It reuses the unchanged F_5 proof clause by clause. This adds a faithful receipt; it is not a new general Balasubramanian–Koblitz theorem. [pcc:unverified]

Frozen original	Actual E/F5, #E=6, r=3; all-k proof retained Issued receipt:176chars; full old written header913chars
Additive wrapper	Same theorem, explicit L and field assumptions Actual rational / chosen-extension / GaloisField maps
Two checks	Kernel proof gate:1 / 0 failures / 0 trusted native Independent official parser:934chars, complete hash match

Mathematical content. For E/F_5: $y^2=x^3+1$, Lean proves nonsingularity, $\#E(F_5)=6$ and the source conditions for r=3, including 3 not dividing 5−1. For every k, all geometric3-torsion points are fixed by Frobenius to the k exactly when 3 divides 5^k−1. An actual order3 point satisfies Frobenius(P)=−P≠P and lies outside the rational point image. For every compatible chosen finite extension K of cardinality5^k, the full geometric3-torsion lies in its actual point-map image exactly under the same divisibility condition. For every positive k an actual GaloisField5k embedding and cardinality are supplied. No rank, determinant, Hasse or pairing conclusion is assumed. [pcc:unverified]

Binding repair. The old issued PCC-20261110-99983d stores a176-character prefix ending inside a named argument; the complete old written header has913characters. Original claim, source and whole37-declaration proof receipt stay unchanged. The new934-character written type includes its field telescope and contains no embedded named-argument assignment or bar delimiter. Its SHA256 is255334b4de88ddc92803431efa042b40aa02db63affd414666c7a20c01cb02ed. An independent native official Lean parser resolves exactly1complete header/0prefix/0unresolved and matches that hash. This procedural check is not proved by the wrapper theorem. [pcc:unverified]

Definitions and proof. ActualPointMaps defines the three maps as Mathlib affine Point.map with explicit parameters. FullBKCertificate applies the frozen actual theorem, retaining all quantifiers, including positive k for the GaloisField clause. Recursive custom-source/config pins supplement the written signature; global semantic type/environment binding and the canonical shared-parser repair remain open. General r≥5, canonical Weil necessity and the complete all-k census are separate remaining BK work. [pcc:unverified]

Checks and reproducibility. New-project whole gate1/0 failures/0 trusted native; printed transitive foundations are propext, Classical.choice and Quot.sound. Three builds, including two failed elaboration states, and all source/config/native protocol/input/output provenance are retained: 44files/25objects, lossless XZ archive8824bytes, with the five-file final gate. Build03 resolves duplicate-instance and conjunction elaboration issues without weakening the target. Lean4.35.0/Mathlib56a00585…, default settings; no holes/custom axioms/native proof evaluation/flags/limits/waivers/new cryptographic experiment/spend/upload. Signed standing remains unverified. [pcc:unverified]

Sources. Original BK1998 Theorem1/conditions/proof pp141–143, earlier full141–145 read receipt credited; source PDF hash ac837b8e…. Frozen ThreeTorsion/QuadraticWitness/EmbeddingCriterion/GaloisFieldCase/BKCertificate and official Mathlib affine Point.map816–850 read. Fresh KB status/two mechanism-negative queries/exact theorem query all127; primary/repository/official fallback. No novelty claim. Source-linked proof map, sources.json and evidence/source-review.json accompany this report. [pcc:unverified]